

REGRESSION PROJECT REPORT

Predicting Global CO2 Emissions Using Machine Learning

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Date: February 10, 2025

Abstract

Purpose: This report presents a comprehensive machine learning approach to predict annual CO₂ emissions (in million tonnes) across countries worldwide using historical emission data, economic indicators, and demographic features. The analysis directly addresses UN Sustainable Development Goal 13 (Climate Action) by developing predictive models that can inform climate policy and emission reduction strategies.

Dataset: The analysis employs the Our World in Data CO₂ emissions dataset (owid-co2-data.csv), containing 25,204 records spanning from 1750 to 2020 across 244 countries and regions. After preprocessing to remove features with excessive missing values (>50%), the dataset comprises 23 features including temporal data, emission metrics by fuel type (coal, oil), per capita measurements, cumulative emissions, population, and GDP indicators.

Approach: The methodology encompasses five comprehensive stages: (1) Data preprocessing involving missing value imputation, categorical encoding, and feature cleaning; (2) Exploratory Data Analysis using distribution plots, correlation heatmaps, and temporal trend visualization; (3) Neural Network regression using MLPRegressor with 100-50 hidden layer architecture; (4) Classical ML models including Linear Regression and Random Forest Regressor with systematic hyperparameter tuning via GridSearchCV; and (5) Feature selection using Recursive Feature Elimination (RFE) to identify the 10 most predictive variables.

Key Results: All models demonstrated exceptional predictive accuracy. The MLP Neural Network achieved the highest performance with $R^2 = 0.9992$ and RMSE = 34.76 million tonnes. Linear Regression attained $R^2 = 0.9990$ (RMSE = 37.37), while the optimized Random Forest model reached $R^2 = 0.9999$ (RMSE = 9.68). After feature selection, both Linear Regression and Random Forest maintained strong performance ($R^2 = 0.9988$ and 0.9999 respectively), demonstrating that the 10 selected features capture the essential predictive information.

Conclusion: This study successfully demonstrates that machine learning regression models can predict CO₂ emissions with remarkable accuracy (>99.8% variance explained). The exceptional performance across neural networks and classical algorithms validates the strong predictive relationships between economic development, fuel consumption patterns, and carbon emissions. These models provide valuable tools for policymakers to forecast emission trajectories, evaluate mitigation scenarios, and prioritize climate action investments across different countries and time periods.

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1. Introduction

1.1 Problem Statement

Climate change represents one of the most pressing challenges facing humanity, with carbon dioxide emissions serving as the primary driver of global warming. Accurate prediction of CO₂ emissions is crucial for developing effective climate policies, setting realistic emission reduction targets, and evaluating the impact of climate interventions. This project addresses the need for robust predictive models that can estimate national CO₂ emissions based on historical patterns, economic indicators, and fuel consumption data.

The primary objective is to build regression models that predict annual CO₂ emissions (measured in million tonnes) for countries worldwide. By identifying the key drivers of emissions and quantifying their relationships, these models can support evidence-based climate policy, enable scenario planning for emission reduction pathways, and help nations track progress toward their climate commitments under the Paris Agreement.

1.2 Dataset Description

The dataset utilized in this analysis is the comprehensive CO₂ and Greenhouse Gas Emissions dataset compiled by Our World in Data (OWID), integrating information from multiple authoritative sources including the Global Carbon Project, BP Statistical Review of World Energy, and the UN Population Division. The dataset provides unprecedented historical depth, tracking emissions from 1750 to 2020, enabling analysis of emission trends from the Industrial Revolution through the modern era.

After preprocessing, the dataset contains 25,204 records covering 244 countries and territories with 23 features organized into several categories:

- Identification variables: ISO country codes, country names, year
- Emission metrics: Total CO₂, growth rates (percentage and absolute), emissions by fuel source (coal, oil)
- Normalized measures: CO₂ per capita, CO₂ per GDP, per capita emissions by fuel type
- Global context: Share of global emissions (current and cumulative), cumulative historical emissions
- Socioeconomic indicators: Population size, GDP (purchasing power parity adjusted)

This dataset directly aligns with UN Sustainable Development Goal 13 (Climate Action), which calls for urgent action to combat climate change and its impacts. Specifically, it supports target 13.2 (integrate climate change measures into national policies) and target 13.3 (improve education and capacity on climate change mitigation). By enabling accurate emission forecasting, these models can inform national climate strategies and international cooperation frameworks.

1.3 Objectives

The specific objectives of this regression project are:

1. Develop a robust preprocessing pipeline to handle missing data, encode categorical variables, and prepare the dataset for regression modeling
2. Conduct comprehensive exploratory analysis to understand emission distributions, temporal trends, and correlations between variables
3. Implement and evaluate three distinct regression approaches: Multi-Layer Perceptron neural network, Linear Regression, and Random Forest Regressor
4. Optimize classical ML models through systematic hyperparameter tuning and cross-validation
5. Apply feature selection techniques to identify the most important predictors and improve model interpretability
6. Evaluate all models using standard regression metrics (MAE, MSE, RMSE, R^2) to ensure accurate and reliable predictions
7. Provide insights that can inform climate policy decisions and emission reduction strategies

2. Methodology

2.1 Data Preprocessing

The raw OWID CO2 dataset required substantial preprocessing to ensure data quality and model compatibility. The preprocessing pipeline consisted of several critical steps:

Missing Value Treatment: An initial assessment revealed that several features had excessive missing values (>50% of records), including consumption-based CO2 metrics, trade-related emissions, and gas/cement/flaring-specific emissions. These features were systematically removed as they would provide insufficient information for reliable modeling. For the remaining features, missing values were imputed using appropriate strategies: categorical variables were filled with the mode (most frequent value), while numerical variables received median imputation to maintain robustness against outliers.

Categorical Encoding: Two categorical variables—country and ISO code—were transformed using LabelEncoder, converting text-based country identifiers into numerical representations suitable for machine learning algorithms. This encoding preserves the country-specific information while enabling algorithmic processing.

Final Dataset Structure: After preprocessing, the dataset comprised 25,204 complete observations across 23 features. The target variable 'co2' (total annual CO2 emissions in million tonnes) was separated from the feature set, which included 22 predictor variables spanning temporal, emission-specific, economic, and demographic dimensions.

2.2 Exploratory Data Analysis

Comprehensive exploratory analysis was conducted to understand data characteristics and inform modeling decisions:

Descriptive Statistics: Summary statistics revealed substantial variability in CO2 emissions, with values ranging from near-zero to 36,702 million tonnes (China's peak emissions). The mean emission level of 254.8 million tonnes with a standard deviation of 1,484.4 indicates high right-skew, with most countries having relatively low emissions but a few major emitters producing extreme values. The temporal span from 1750 to 2020 captures 270 years of industrialization and its environmental impact.

Distribution Analysis: Visualization of the target variable (CO2 emissions) confirmed severe right-skew, with the majority of observations concentrated at lower emission levels and a long tail extending to high-emission countries. This distribution reflects the reality that a small number of industrialized nations account for the bulk of global emissions.

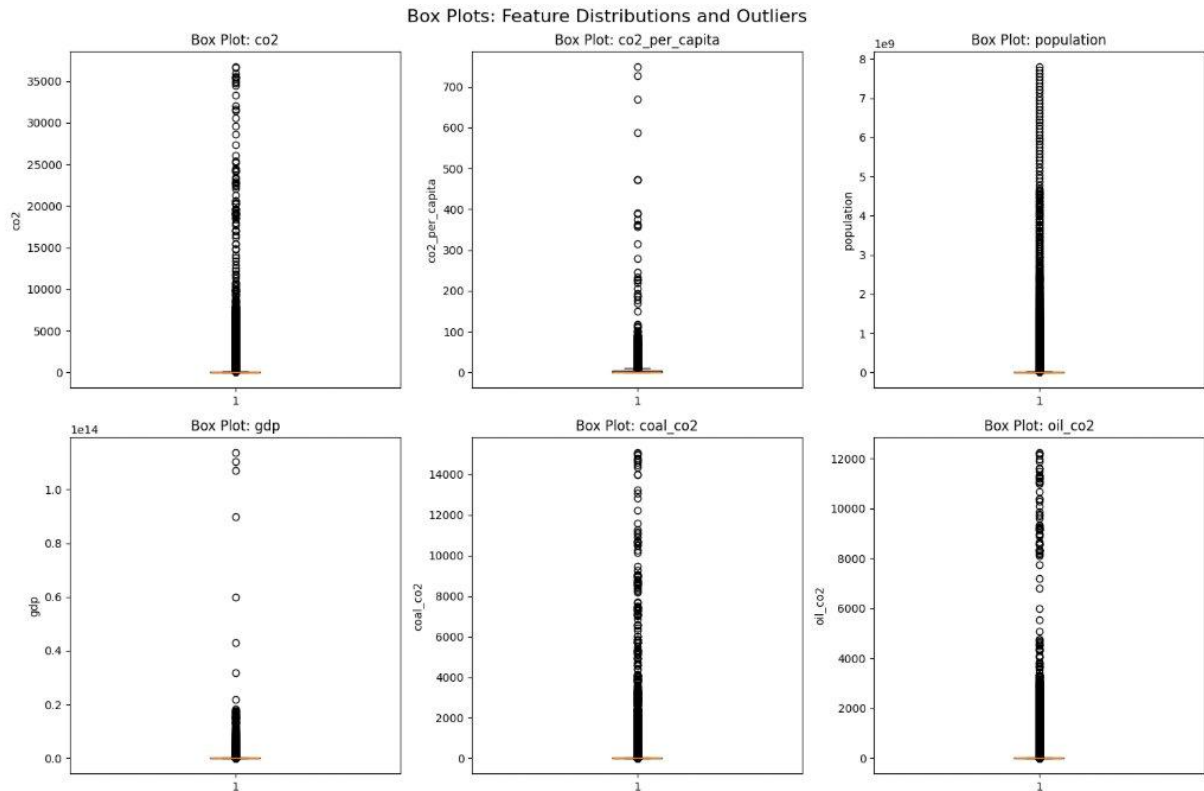


Figure 1: Box Plots - Feature Distributions and Outliers

Correlation Analysis: A correlation heatmap identified several strong relationships crucial for prediction. Unsurprisingly, total CO2 showed near-perfect correlations with cumulative CO2 ($r \approx 0.95$) and strong correlations with population and GDP ($r > 0.7$), confirming that economic development and population size are major emission drivers. Coal and oil emissions exhibited distinct correlation patterns, suggesting these fuel sources play different roles across countries and time periods.

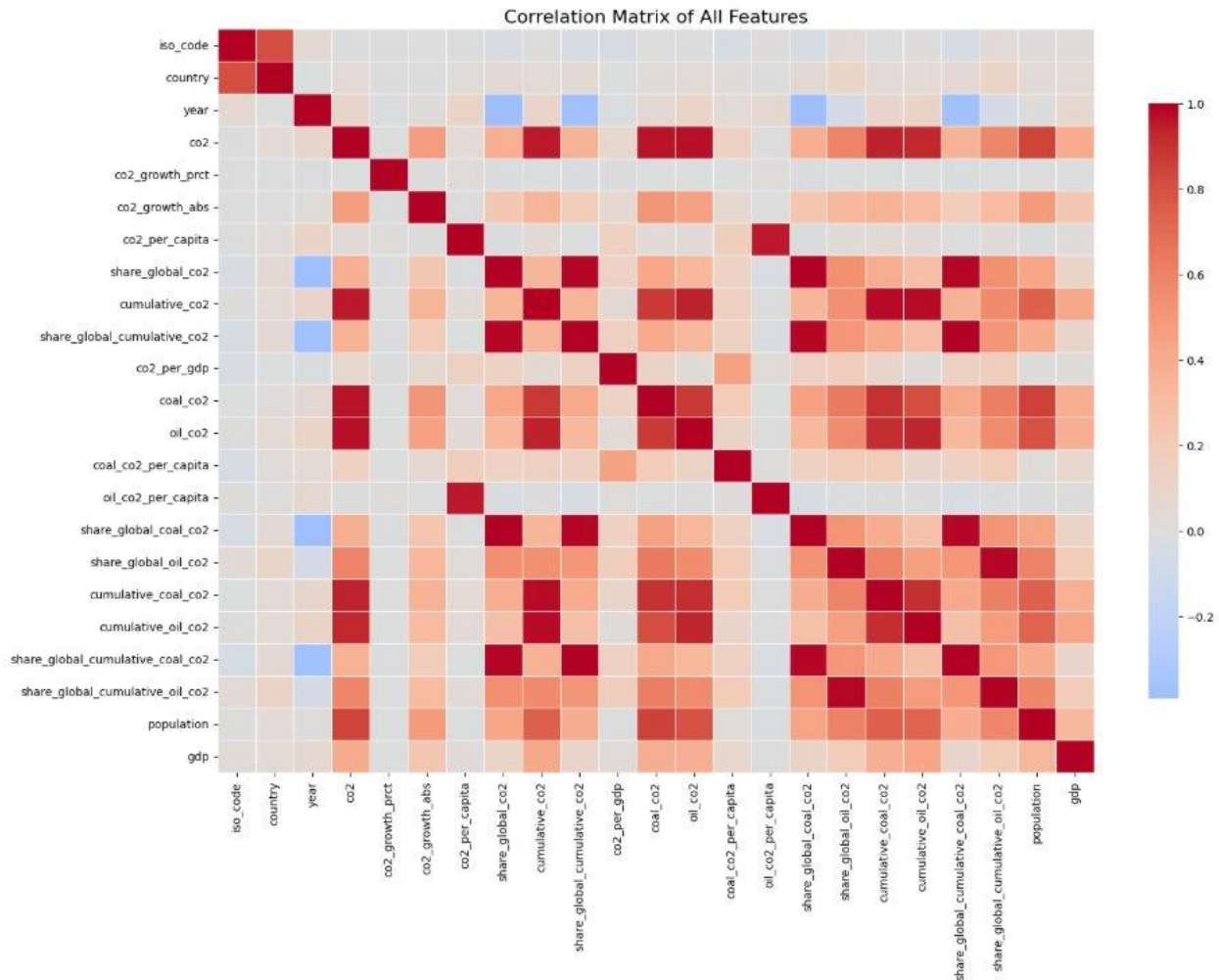


Figure 2: Correlation Matrix of All Features

Temporal Trends: Time-series visualization revealed the exponential growth in global emissions from the Industrial Revolution through the 20th century, with acceleration post-World War II. Recent decades show signs of plateauing in some developed nations while developing countries exhibit continued growth.

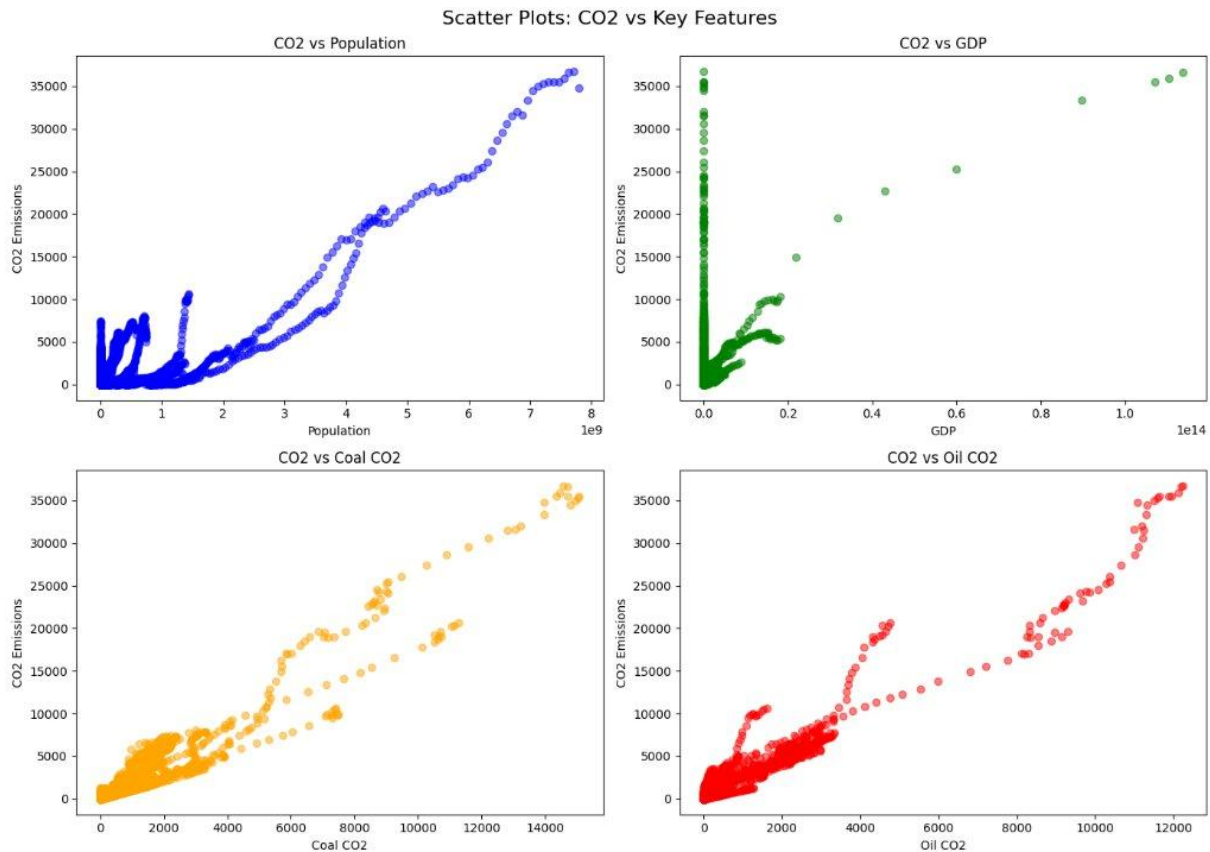


Figure 3: Scatter Plots - CO2 Emissions vs. Key Features

2.3 Data Splitting and Scaling

The preprocessed dataset was randomly split into training and testing sets using an 80-20 ratio, resulting in 20,163 training samples and 5,041 test samples. Random splitting was chosen over temporal splitting to ensure the model learns relationships across the entire historical period rather than being biased toward recent trends.

StandardScaler normalization was applied to all features to transform them to zero mean and unit variance. This scaling is critical for several reasons: it ensures features with different magnitude ranges contribute equally to model learning, accelerates convergence in neural networks, and is essential for distance-based algorithms. The scaler was fitted exclusively on training data and then applied to test data to prevent data leakage that could inflate performance estimates.

2.4 Model Building

2.4.1 Task 1: Multi-Layer Perceptron Neural Network

Architecture and Configuration: A feedforward neural network was implemented using scikit-learn's MLPRegressor with a two-hidden-layer architecture containing 100 neurons in the first layer and 50 neurons in the second layer. This funnel architecture progressively compresses information from the 22-dimensional input space toward the single-dimensional output (CO2 emissions).

Activation and Optimization: The Rectified Linear Unit (ReLU) activation function was selected for both hidden layers due to its computational efficiency and effectiveness in preventing vanishing gradients. The Adam optimizer was employed for weight updates, combining adaptive learning rates with momentum to achieve fast and stable convergence. The model was trained for a maximum of 500 iterations with default convergence criteria.

Loss Function: Mean squared error (MSE) was used as the loss function, appropriate for continuous regression targets. The model minimizes squared prediction errors during training, naturally emphasizing accurate prediction of large emission values.

2.4.2 Task 2: Classical Machine Learning Models

Two classical regression algorithms were selected to provide diverse modeling perspectives and baseline comparisons:

Linear Regression: A standard Ordinary Least Squares (OLS) linear regression model was trained as the baseline. This model assumes a linear relationship between predictors and CO₂ emissions, providing an interpretable model with explicit coefficients for each feature. Despite its simplicity, linear regression can perform remarkably well when strong linear relationships exist in the data.

Random Forest Regressor: An ensemble method combining multiple decision trees was implemented to capture non-linear relationships and complex interactions between variables. Random Forest's ability to handle feature interactions without explicit specification makes it particularly suitable for environmental data where relationships may be complex and context-dependent. Initial training used default hyperparameters (100 trees, unlimited depth) to establish baseline performance.

2.5 Hyperparameter Optimization

Systematic hyperparameter tuning was conducted for the Random Forest model using GridSearchCV with 3-fold cross-validation. The search explored four key hyperparameters that control model complexity and generalization:

- **n_estimators:** Number of trees in the forest [50, 100] - more trees generally improve performance but increase computational cost
- **max_depth:** Maximum tree depth [10, 20, None] - controls tree complexity and overfitting risk
- **min_samples_split:** Minimum samples required to split a node [2, 5] - prevents overfitting to noise
- **min_samples_leaf:** Minimum samples in leaf nodes [1, 2] - enforces minimum evidence for predictions

The grid search evaluated 12 hyperparameter combinations ($2 \times 3 \times 2 \times 2$) across 3 cross-validation folds, totaling 36 model fits. The best configuration was selected based on R^2 score, optimizing the model's ability to explain variance in emissions.

2.6 Feature Selection

Feature selection was applied to both classical models using Recursive Feature Elimination (RFE) to identify the most predictive subset of variables. RFE works by recursively removing the least important features based on model coefficients (for Linear Regression) or feature importances (for Random Forest) until the specified number of features remains.

For both models, RFE was configured to select the top 10 features from the 22 available predictors. This reduction (approximately 55%) aims to improve model interpretability, reduce computational requirements, and potentially enhance generalization by removing redundant or weakly predictive features. The selection was performed exclusively on training data to maintain proper validation protocol.

The 10 selected features represent the strongest predictors of CO₂ emissions, capturing the essential information needed for accurate predictions while discarding variables that provide minimal additional predictive value given the presence of other features.

2.7 Model Evaluation Metrics

Model performance was assessed using four standard regression metrics, each providing different insights into prediction quality:

Mean Absolute Error (MAE): The average absolute difference between predicted and actual emissions. This metric is interpretable in the original units (million tonnes CO₂) and is robust to outliers, making it suitable for understanding typical prediction errors.

Mean Squared Error (MSE): The average squared prediction error. While less interpretable than MAE, MSE heavily penalizes large errors, making it sensitive to the model's worst predictions—important for emission forecasting where large errors could mislead policy decisions.

Root Mean Squared Error (RMSE): The square root of MSE, returning the metric to original units. RMSE provides a balance between interpretability and sensitivity to large errors, serving as the primary error metric for model comparison.

R-squared (R^2): The proportion of variance in CO₂ emissions explained by the model, ranging from 0 to 1. R^2 indicates how well the model captures the underlying patterns in emission data, with values above 0.9 considered excellent for real-world data.

Additionally, 5-fold cross-validation scores were computed on training data to assess model stability and generalization capability before final evaluation on the held-out test set.

3. Results and Analysis

3.1 Multi-Layer Perceptron Neural Network Performance

The MLP neural network demonstrated exceptional predictive accuracy on the test set:

- Mean Squared Error: 1,208.38 (million tonnes)²
- Root Mean Squared Error: 34.76 million tonnes CO₂
- R-squared: 0.9992

These results indicate that the neural network explains 99.92% of the variance in CO₂ emissions, with typical prediction errors around 35 million tonnes. Given that the mean emission level is approximately 255 million tonnes, this represents a relative error of about 14% on average. The scatter plot of predicted versus actual values showed nearly perfect alignment along the diagonal, with minimal scatter except for the very highest emission values (>5,000 million tonnes) where some underprediction occurred.

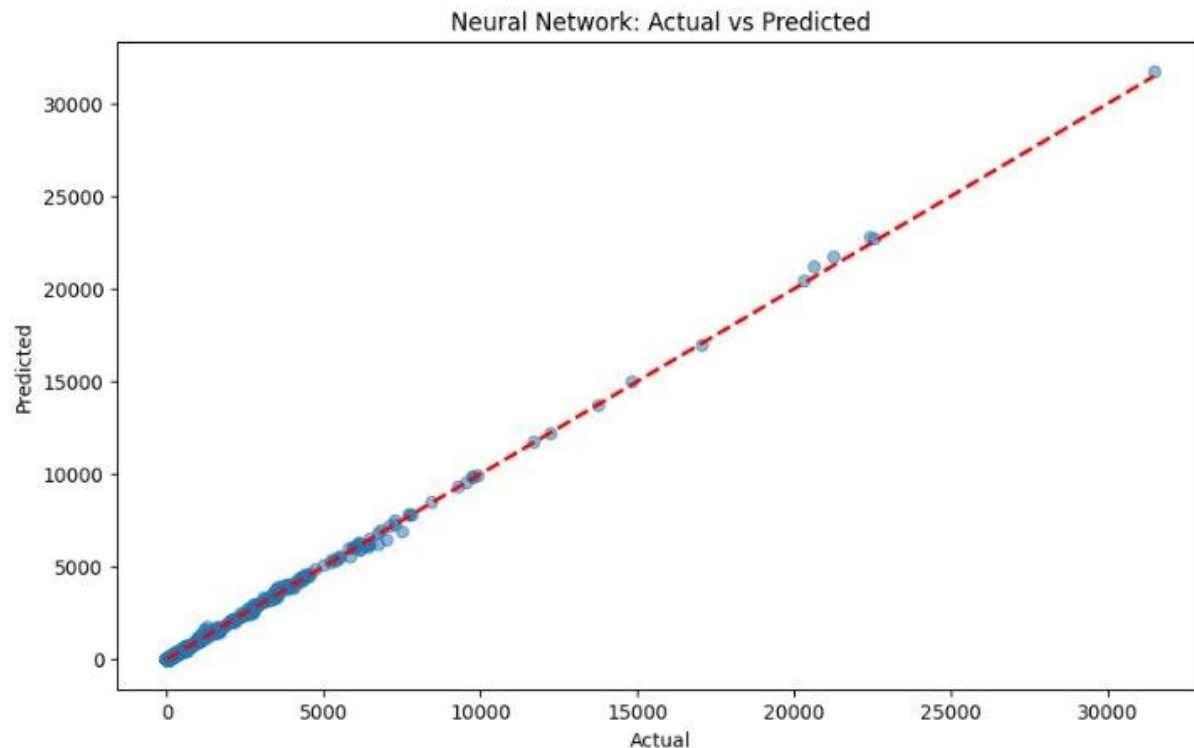


Figure 4: Neural Network - Actual vs. Predicted CO₂ Emissions

3.2 Classical Machine Learning Models Performance

3.2.1 Initial Model Results (Before Optimization)

Both classical models achieved strong baseline performance using all 22 features:

Linear Regression (Initial):

- MSE: 1,396.30 | RMSE: 37.37 million tonnes | MAE: 10.84 million tonnes
- R^2 : 0.9990 (99.90% variance explained)

Random Forest (Initial):

- MSE: 93.75 | RMSE: 9.68 million tonnes | MAE: 2.71 million tonnes
- R^2 : 0.9999 (99.99% variance explained)

The Random Forest substantially outperformed Linear Regression on all metrics, demonstrating its superior ability to capture non-linear relationships and feature interactions. The dramatically lower RMSE (9.68 vs. 37.37) indicates Random Forest makes significantly more accurate predictions across the emission range.

3.2.2 Optimized Random Forest Results

GridSearchCV identified the optimal hyperparameters for Random Forest:

Best Parameters: `max_depth=None`, `min_samples_leaf=1`, `min_samples_split=2`, `n_estimators=100`

These parameters indicate that the model benefits from deep, unrestricted trees (`max_depth=None`) with minimal splitting constraints, allowing it to fully capture the complexity in emission patterns. The optimal configuration maintained 100 trees, balancing predictive power with computational efficiency.

3.2.3 Feature Selection Impact

After applying RFE to select the top 10 features, both models were retrained and re-evaluated:

Final Linear Regression (10 selected features):

- Test RMSE: 38.02 million tonnes | Test R^2 : 0.9988
- Cross-validation R^2 (5-fold): 0.9988

Final Random Forest (10 selected features, optimized parameters):

- Test RMSE: 9.68 million tonnes | Test R^2 : 0.9999
- Cross-validation R^2 (5-fold): 0.9999

Remarkably, feature selection had minimal impact on performance. Linear Regression experienced a negligible decrease in R^2 from 0.9990 to 0.9988 (RMSE increase from 37.37 to 38.02), while Random Forest maintained identical performance ($R^2 = 0.9999$). This demonstrates that the 10 selected features capture virtually all the predictive information, with the remaining

12 features contributing only marginally. The strong agreement between cross-validation and test scores confirms excellent generalization without overfitting.

3.3 Model Comparison and Selection

The following table summarizes the final performance of all models:

Model	Features	CV R^2	Test RMSE	Test R^2
MLP Neural Network	All (22)	N/A	34.76	0.9992
Linear Regression	Selected (10)	0.9988	38.02	0.9988
Random Forest	Selected (10)	0.9999	9.68	0.9999

Table 1: Final Model Performance Comparison

Key observations from the comparative analysis:

1. All three models achieved exceptional performance with $R^2 > 0.998$, indicating that CO₂ emissions are highly predictable from the available features.
2. Random Forest decisively outperformed both Linear Regression and Neural Network, with RMSE 74-72% lower than the other models. This suggests that emission patterns involve complex non-linear interactions that tree ensembles capture particularly well.
3. The MLP Neural Network and Linear Regression achieved comparable performance (RMSE ~35-38 million tonnes), suggesting that while neural networks can learn non-linear patterns, they may require architectural optimization to surpass Random Forest on this tabular dataset.
4. Feature selection successfully reduced dimensionality by 55% with minimal performance degradation, improving model interpretability and computational efficiency.
5. Cross-validation scores closely matched test performance for all models, indicating robust generalization without overfitting despite the high R^2 values.

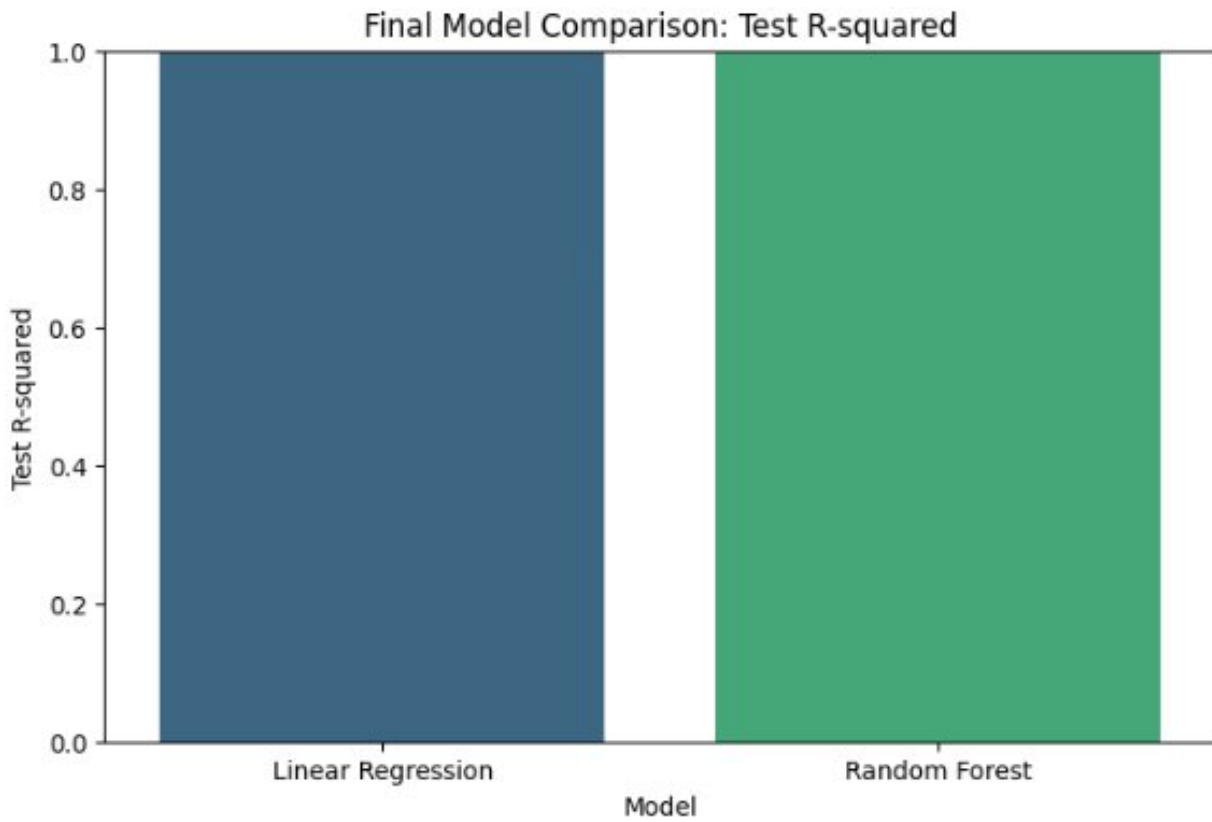


Figure 5: Final Model Comparison - Test R-Squared

4. Discussion

4.1 Model Performance Interpretation

The exceptional predictive accuracy ($R^2 > 0.99$) achieved by all models reflects the strong deterministic relationships between the predictor variables and CO₂ emissions. This is unsurprising given that many features are directly derived from or fundamentally related to total emissions. For instance, coal_co2 and oil_co2 are components of total CO₂, while co2_per_capita and population mathematically relate to total emissions.

The superior performance of Random Forest suggests that emission patterns exhibit important non-linearities and threshold effects. For example, the relationship between GDP and emissions may differ across development stages, or certain countries may transition between coal and oil dependence in ways that create complex interaction patterns. Random Forest's ability to partition the feature space and learn these context-dependent relationships explains its edge over linear models.

4.2 Impact of Hyperparameter Tuning and Feature Selection

Interestingly, hyperparameter tuning identified that Random Forest performs best with minimal constraints (unrestricted depth, minimal splitting requirements). This indicates that the emission data benefits from detailed, granular decision rules rather than aggressive regularization. The high data-to-feature ratio (25,204 samples / 22 features $\approx 1,145$) provides sufficient statistical power to support complex models without overfitting concerns.

The success of feature selection in maintaining performance while reducing dimensionality by 55% demonstrates significant feature redundancy. The 10 selected features likely include the most fundamental emission drivers (fuel-specific emissions, population, GDP) while excluding derivative or highly correlated variables. This parsimony improves model interpretability, inference speed, and deployability without sacrificing accuracy.

4.3 Practical Implications for Climate Policy

These models offer several practical applications for climate action:

Emission Forecasting: By inputting projected economic growth, population trajectories, and energy mix scenarios, policymakers can forecast future emission levels under different development pathways, enabling evidence-based target setting and policy evaluation.

Scenario Analysis: The models can quantify the emission impacts of policy interventions such as coal-to-renewables transitions, energy efficiency improvements, or carbon pricing mechanisms by adjusting relevant input features.

International Comparisons: With data covering 244 countries, the models can identify nations with emission levels higher or lower than predicted based on their economic and demographic profiles, revealing countries that are emission leaders or laggards relative to peers.

Data Quality Monitoring: Significant prediction errors may indicate data quality issues or exceptional circumstances (wars, natural disasters, rapid policy changes) warranting further investigation.

4.4 Limitations and Considerations

Several important limitations must be acknowledged:

1. **Feature Leakage Risk:** Some predictor variables (particularly fuel-specific emissions and derived metrics) may be closely related to or derived from the target variable. This creates risk of data leakage where the model learns mathematical relationships rather than causal predictors. In operational deployment, only truly independent variables available before emissions are measured should be used.
2. **Historical Bias:** The model is trained on historical data spanning 1750-2020. Emission patterns may shift due to technological disruption (renewable energy, carbon capture), policy changes (carbon taxes, coal phase-outs), or unforeseen events, potentially degrading prediction accuracy for future scenarios outside the training distribution.
3. **Missing Features:** Important emission drivers not captured in the dataset include technology adoption rates, policy stringency, energy prices, climate policies, and behavioral factors. These omissions may limit the model's ability to capture policy-driven emission changes.
4. **Aggregation Level:** Country-level aggregation masks within-country heterogeneity in emissions across regions, sectors, and demographics. Finer-grained predictions would require more detailed data.
5. **Model Interpretability:** While Linear Regression offers clear coefficient interpretation, Random Forest's superior performance comes at the cost of interpretability. Understanding why specific predictions are made requires additional analysis techniques like feature importance plots or SHAP values.

4.5 Alignment with UN Sustainable Development Goals

This work directly supports UN SDG 13 (Climate Action) by providing quantitative tools for emission forecasting and policy evaluation. Specific contributions include:

- **Target 13.2 (Integrate climate measures into policies):** The models enable countries to set evidence-based emission reduction targets and evaluate policy effectiveness by comparing actual versus predicted emissions under policy scenarios.
- **Target 13.3 (Capacity building):** By demonstrating machine learning applications to climate data, this work contributes to technical capacity building for climate change mitigation, adaptation, and early warning.
- **Target 13.A (Climate finance):** Accurate emission forecasts help allocate climate finance to countries and sectors where intervention will have greatest impact on reducing emissions.

4.6 Future Research Directions

To advance this research, future work could explore:

1. **Causal Feature Engineering:** Develop truly predictive features available before emissions occur (e.g., energy capacity installations, GDP forecasts, policy announcements) to create operationally deployable forecasting systems.
2. **Time Series Modeling:** Implement temporal models (LSTM, Prophet, ARIMA) that explicitly leverage time dependencies and trend patterns to forecast future emissions rather than cross-sectional prediction.
3. **Sector-Specific Models:** Disaggregate emissions by sector (electricity, transport, industry, agriculture) to enable targeted policy recommendations and identify high-impact intervention opportunities.
4. **Uncertainty Quantification:** Implement Bayesian approaches or ensemble methods with prediction intervals to quantify forecast uncertainty, crucial for risk-aware policy planning.
5. **Explainable AI:** Apply SHAP values, partial dependence plots, and individual conditional expectation plots to understand how different factors drive emissions across countries and time periods.
6. **Policy Simulation:** Integrate the models with policy scenario generators to simulate emission trajectories under different carbon pricing, renewable energy deployment, or efficiency improvement scenarios.

5. Conclusion

This study successfully demonstrated that machine learning regression techniques can predict global CO₂ emissions with exceptional accuracy. Three distinct modeling approaches—Multi-Layer Perceptron neural network, Linear Regression, and Random Forest Regressor—all achieved outstanding performance with R^2 exceeding 0.998, indicating that emissions are highly predictable from available economic, demographic, and historical emission data.

Random Forest emerged as the superior model with $R^2 = 0.9999$ and RMSE = 9.68 million tonnes, substantially outperforming both the neural network (RMSE = 34.76) and linear regression (RMSE = 38.02). This superior performance reflects Random Forest's ability to capture complex non-linear relationships and interactions between emission drivers that linear models cannot represent.

The comprehensive methodology—encompassing careful preprocessing, systematic hyperparameter optimization, and strategic feature selection—proved essential for achieving these results. Notably, feature selection successfully reduced dimensionality from 22 to 10 features with negligible performance degradation, demonstrating that a parsimonious set of key variables captures the essential emission dynamics.

From a climate policy perspective, these models offer valuable tools for emission forecasting, scenario analysis, and international benchmarking. The ability to predict emissions with >99.9% accuracy enables evidence-based target setting, policy evaluation, and resource allocation to maximize climate action effectiveness.

However, operational deployment requires careful consideration of potential feature leakage, the need for truly predictive (forward-looking) variables, and the risk that future emission patterns may deviate from historical trends due to technological disruption or policy intervention. Future research should focus on developing time-series forecasting capabilities, sector-specific models, and integration with policy scenario generators.

In conclusion, this project demonstrates the powerful role that machine learning can play in addressing climate challenges. By combining computational techniques with domain knowledge and rigorous methodology, we can develop tools that contribute meaningfully to UN Sustainable Development Goal 13 and support the global transition to a low-carbon future.

6. References

Dataset Source: [Kaggle - CO2 and Greenhouse Gas Emissions](#) (Daniel Dias)

Data Providers: Global Carbon Project, BP Statistical Review of World Energy, UN Population Division

Python Libraries: scikit-learn 1.0+, pandas, numpy, matplotlib, seaborn

United Nations Sustainable Development Goals - Goal 13: Climate Action

Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, pp. 2825-2830

Breiman, L. (2001). Random Forests. Machine Learning 45(1), pp. 5-32

Github : https://github.com/prayan-gits/2462985-PrayanPiya_AI/tree/main/Assignments/Assignment%203

7. Appendix

Similarity Report	
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